

● HARD

● GEOMETRY AND TRIGONOMETRY

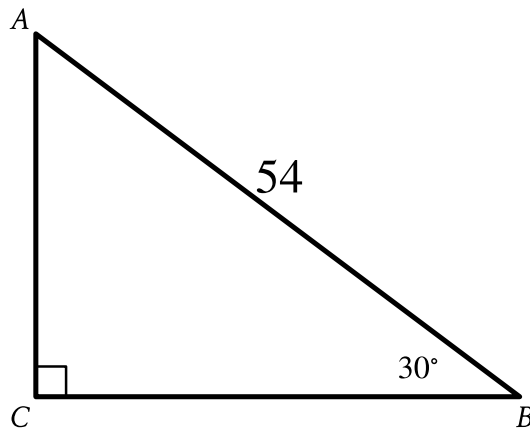
Geometry and Trigonometry

12

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- The length of side upper A upper B is 54.
- The measure of angle upper B is 30° .
- Angle upper C is a right angle.
- A note indicates the figure is not drawn to scale.

Right triangle ABC is shown. What is the value of $\tan A$?

- A. $\frac{\sqrt{3}}{54}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. $27\sqrt{3}$

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

- A. 60π
- B. 120π
- C. 600π
- D. $2,400\pi$

Q3

GRID-IN

GEOMETRY AND TRIGONOMETRY

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy -plane is a circle. What is the length of the circle's radius?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY

The perimeter of an equilateral triangle is 852 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GEOMETRY AND TRIGONOMETRY

Points Q and R lie on a circle with center P . The radius of this circle is 9 inches. Triangle PQR has a perimeter of 31 inches. What is the length, in inches, of $\overline{QR}$?

- A. $13\sqrt{2}$
- B. 13
- C. $9\sqrt{2}$
- D. 9

Q6

GEOMETRY AND TRIGONOMETRY

In triangle XYZ , the measure of angle X is 90° . Point W lies on segment YZ , and segment WX is perpendicular to segment YZ . The length of segment WY is 572, and the length of segment WX is 429. What is the value of $\tan Z$?

- A. $\frac{3}{5}$
- B. $\frac{3}{4}$
- C. $\frac{4}{5}$
- D. $\frac{4}{3}$

A circle has center O , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 3 inches. What is the circumference, in inches, of the circle?

- A. 3
- B. 6
- C. 9
- D. 24

Q8

GRID-IN

GEOMETRY AND TRIGONOMETRY

The length of a rectangle's diagonal is $3\sqrt{17}$, and the length of the rectangle's shorter side is 3. What is the length of the rectangle's longer side?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x^2 + 14x + y^2 = 6y + 109$$

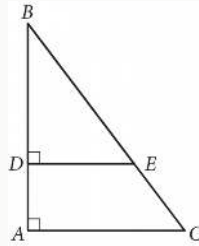
In the xy -plane, the graph of the given equation is a circle. What is the length of the circle's radius?

- A. $\sqrt{109}$
- B. $\sqrt{149}$
- C. $\sqrt{167}$
- D. $\sqrt{341}$

Q10

GRID-IN

GEOMETRY AND TRIGONOMETRY



In the figure above, $\tan B = \frac{3}{4}$. If $BC = 15$ and $DA = 4$, what is the length of $\overline{DE}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.